

RAHUL MITRA

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EDUCATION

National University of Singapore Bachelor of Engineering, Industrial Systems Engineering - Pursuing a Second Major in Computer Science and a Minor in Mathematics. - Awarded Certificate of Outstanding Performance in 3D Computer Vision (CS4277): top student in a class of 24, NUS School of Computing.	Aug 2023 – Jul 2027
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WORK EXPERIENCE

Operations Research (NUS System Design Project), Contract, STMicroelectronics Design an AI-driven put-away recommendation system for STMicroelectronics' Singapore warehouse, replacing experience-based slotting with capacity-aware decisions drawn from 3 live inputs: picking history, demand forecasts, and available capacity.	Aug 2026 – Present
Robotics Vision Engineer, Amazon (BlendED AI+X) Build vision systems for 3 imaging workstreams: camera ISP enhancement, super-resolution, and image restoration under motion and low light.	Jul 2026 – Present
Operational AI Systems & Data Engineer (Contract), Abbott Laboratories Assemble a 5-stage offline process-intelligence platform for globally distributed spray-drying plants: GPU OCR and layout parsing, recursive chunking, ONNX embeddings, a Chroma vector store, and LangChain over a locally hosted Gemma model.	Jun 2026 – Present
Software Engineer (Citizen Developer), People's Association - Delivered a digital garden-plot balloting platform across admin and public frontends, backend, GIS mapping, and deployment, now production-ready for national rollout across Singapore. - Integrated Singpass Login (OIDC) to secure resident sign-in and Myinfo to verify identity and address, keeping garden-plot balloting fair for 150 residents signed up at release.	Sep 2025 – Sep 2026
Data Analytics & Product Management Intern, Abbott Laboratories - Developed a SimPy discrete-event digital twin and OR-Tools CP-SAT solver for hybrid flow-shop scheduling, then compared 3 approaches (heuristic, MIP, and genetic algorithm) to turn production objectives and constraints into decision-ready schedules. - Productionized a team-built APC simulator for live internal users on Azure App Service with Docker packaging, access restrictions, and runtime and session hardening.	Jan 2026 – Jun 2026

PROJECTS

Hackathon Build, OnTheSpectrum: AI Engineer Hackathon 2026 Created a game of AI-driven infinite levels in a 7-hour hackathon build, wiring Codex and Blender MCP into a local-first Blender-to-Three.js pipeline for GLB assets, previews, and playable-world QA.	May 2026
Personal Project, Hailo Edge AI Training and Deployment Trained YOLO26 through transfer learning and reshaped its architecture with the Hailo Dataflow Compiler, running document detection on a Hailo-8L rated 13 TOPS at 1.5 W to speed up large-batch scanning.	Mar 2026
Finalist, Volt Pulse SG: SMU Hack For Cities (Top 8 Finalist of 50+ teams) Integrated SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning into an autonomous loop managing electricity and utility savings across multiple HDB households at once, placing top 8 of 50+ teams.	Jan 2026
Hackathon Project, Waaah Comics: 24-Hour Gesture-to-Comic Build Turned MediaPipe 3D hand landmarks into generated comic strips in a 24-hour sprint with Gemini 2.0 Flash, Veo 3, and Konva.js canvas tooling.	Jan 2026

LEADERSHIP AND ACTIVITIES

NTUC Health Active Ageing Centre, Elderly Care Volunteer Coordinate weekly senior-engagement activities, building empathy for healthcare, accessibility, and community-focused product design.	Sep 2025 – Sep 2026
Republic of Singapore Navy, Base Support Assistant Compressed a 6-month food-waste analysis cycle into a 3-minute reporting process with an Excel VBA automation workflow inside a security-constrained environment.	Jan 2022 – Jan 2023

SKILLS AND CERTIFICATIONS

Technical: Python, TypeScript, React, Next.js, Three.js, FastAPI, PyTorch, ONNX, Docker, CI/CD, Azure; 3D CV: camera models, epipolar geometry, absolute pose (PnP), SfM, bundle adjustment, multi-view stereo
Certifications: NVIDIA Disaster Risk Monitoring with Satellite Imagery; Deep Reinforcement Learning Hands-On (Packt)